Lab Report No 11
Demonstration of Matrix Keyboard Control using 8051 Interrupts.

Embedded Systems

Submitted By:

[Huzaifa Shahbaz]

[21-CE-009]

[Aqsa Shah]

[21-CE-039]

Submitted to:

Engr. Bushra Fiaz

Dated:

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Department of Computer Engineering,

HITEC University, Taxila

Lab Task No 2:

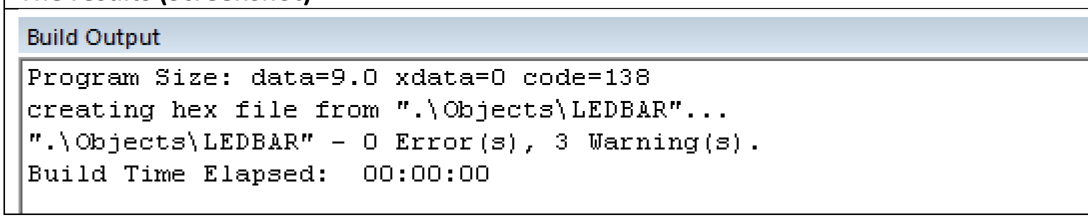
Solution:

Brief description (3-5 lines)
Write a program which generate the following patterns on LED bar. a) If button 0 is pressed =>; Turn on LED bar from left to right b) If button 1 is pressed =>; Turn on LED bar from right to left
The code

```
ORG 0000H
LJMP MAIN
ORG 0030H
MAIN:
   MOV P1, #0FFH      ; Configure P1 as input for buttons
   MOV P2, #00H       ; Configure P2 as output for LED bar
CHECK_BUTTONS:
   MOV A, P1          ; Read button states
   ; Check if button 0 is pressed (active low)
   JNB P1.0, LEFT_TO_RIGHT
   ; Check if button 1 is pressed (active low)
   JNB P1.1, RIGHT_TO_LEFT
   SJMP CHECK_BUTTONS
LEFT_TO_RIGHT:
   MOV R0, #8         ; Counter for 8 LEDs
   MOV R1, #80H       ; Start with leftmost LED (10000000B)
L2R_LOOP:
   MOV P2, R1         ; Output pattern to LED bar
   LCALL DELAY        ; Add delay
   MOV A, R1          ; Move current pattern to accumulator
   RR A               ; Rotate right
   MOV R1, A          ; Store back to R1
   DJNZ R0, L2R_LOOP  ; Repeat until all LEDs are lit
   MOV P2, #0FFH      ; Turn on all LEDs
   LCALL DELAY
   MOV P2, #00H       ; Turn off all LEDs
```

```
    SJMP CHECK_BUTTONS
RIGHT_TO_LEFT:
    MOV R0, #8          ; Counter for 8 LEDs
    MOV R1, #01H        ; Start with rightmost LED (00000001B)
R2L_LOOP:
    MOV P2, R1          ; Output pattern to LED bar
    LCALL DELAY         ; Add delay
    MOV A, R1           ; Move current pattern to accumulator
    RL A                ; Rotate left
    MOV R1, A           ; Store back to R1
    DJNZ R0, R2L_LOOP   ; Repeat until all LEDs are lit
    MOV P2, #0FFH       ; Turn on all LEDs
    LCALL DELAY
    MOV P2, #00H        ; Turn off all LEDs
    SJMP CHECK_BUTTONS
DELAY:                  ; Delay subroutine
    MOV R6, #5
DELAY_OUTER:
    MOV R7, #200
DELAY_INNER:
    DJNZ R7, DELAY_INNER
    DJNZ R6, DELAY_OUTER
    RET
END
```

The results (Screenshot)



Build Output

```
Program Size: data=9.0 xdata=0 code=138
creating hex file from ".\Objects\LEDBAR"...
".\Objects\LEDBAR" - 0 Error(s), 3 Warning(s).
Build Time Elapsed:  00:00:00
```

Conclusion

In conclusion, the 8051 microcontroller was used in a demonstration of interrupt-based matrix keyboard control, demonstrating its efficiency and reliability. Key presses were detected through hardware interrupts, eliminating the need for continuous polling. The matrix keyboard interface was reliable for user input, and the interrupt service routine effectively handled key scanning and decoding, proving the power of interrupt-driven I/O.